

High-Dimensional Instrumental Variable Quantile Regression under Weak Instruments: A Monte Carlo Study

This thesis studies how IVQR-based estimators behave when two problems occur jointly:

1. the number of controls is large relative to sample size;
2. the instruments are weak or moderately weak.

The objective is to compare the finite-sample performance of several IVQR-type estimators under controlled Monte Carlo designs with high-dimensional controls and varying instrument strength.

- a. Full-control IVQR - uses all controls in the IVQR/GMM-IVQR estimation.
- b. Post-selection IVQR - selects controls first, then runs IVQR.
- c. DML-IVQR - uses penalized quantile regression and orthogonalized moments.

The central estimator is DML-IVQR, which uses Neyman-orthogonal moment conditions and regularization to reduce sensitivity to nuisance-parameter estimation.

1. The Research Question

How reliable is DML-IVQR for estimating structural quantile treatment effects under high-dimensional controls and weak instruments?

This general question is divided into four more specific questions:

1. How does IVQR perform when the number of controls is large relative to the sample size?
2. At what level of instrument strength does IVQR inference begin to break down?
3. Does using regularization improve IVQR performance compared with using all controls?
4. Are weak-instrument problems more severe at tail quantiles than at the median?

2. The Model.

We use a partially linear structural IVQR framework. The main object of interest is the structural quantile treatment effect: $\alpha(\tau)$, $\tau \in \{0.25, 0.50, 0.75\}$.

The structural quantile function is written as:

$$q_\tau(D, W) = \alpha(\tau)D + g_\tau(W),$$

where $\alpha(\tau)$ is the target structural quantile treatment effect and $g_\tau(W)$ is an unknown nuisance function.

The nuisance function $g_\tau(W)$ is approximated by a high-dimensional dictionary of technical controls $g_\tau(W) \approx X'\beta(\tau)$, where $X = h(W)$. Working IVQR specification becomes:

$$q_\tau(D, W) \approx \alpha(\tau)D + X'\beta(\tau).$$

IVQR Moment Condition

Identification is based on the IVQR conditional moment restriction

$$P[Y \leq D\alpha(\tau) + X'\beta(\tau) \mid X, Z] = \tau.$$

Equivalently,

$$E[\tau - \mathbf{1}\{Y \leq D\alpha(\tau) + X'\beta(\tau)\} \mid X, Z] = 0.$$

For estimation, this conditional moment restriction can be transformed into unconditional moments:

$$E[(\tau - \mathbf{1}\{Y \leq D\alpha(\tau) + X'\beta(\tau)\}) \Psi(X, Z)] = 0,$$

where $\Psi(X, Z)$ is a vector of instruments and functions of the controls.

The low-dimensional parameter of interest is $\alpha(\tau)$, while $\beta(\tau)$ is a high-dimensional nuisance parameter.

Endogeneity

The treatment variable D is endogenous. This means that D is correlated with unobserved determinants of the outcome Y . Because of this endogeneity, ordinary quantile regression is generally not valid for estimating the structural quantile treatment effect $\alpha(\tau)$.

The instrumental variable Z is used to address this problem. The instrument must affect the endogenous treatment D , but it should not directly affect the outcome Y except through D . The causal structure can be summarized as:

$$Z \rightarrow D \rightarrow Y.$$

The excluded direct path

$$Z \rightarrow Y$$

is ruled out by the instrumental variable assumption.

Weak-Instrument Design

In the Monte Carlo design, instrument strength is controlled through the first-stage equation

$$D = \pi Z + X'\gamma + v,$$

where $\pi \in \{1.0, 0.5, 0.25, 0.10\}$ measures the strength of the instrument.

As π decreases, the relationship between Z and D becomes weaker. This allows the thesis to study when IVQR-based estimation and inference begin to deteriorate.

3. Estimators

We compare three IVQR-based estimators:

1. Full-control IVQR;
2. Post-selection IVQR;
3. DML-IVQR.

All three estimators target the same structural quantile treatment effect,

$$\alpha(\tau), \quad \tau \in \{0.25, 0.50, 0.75\}.$$

They differ in how they handle the high-dimensional nuisance component $X'\beta(\tau)$.

Full-Control IVQR

The Full-control IVQR estimator uses the complete set of technical controls X in the IVQR estimation. For each quantile τ , the working model is

$$q_\tau(D, W) \approx \alpha(\tau)D + X'\beta(\tau).$$

The estimator solves the sample analogue of the IVQR moment condition using all available controls:

$$\hat{\alpha}_{Full}(\tau) = \arg \min_{a \in \mathcal{A}} \hat{g}_n(a)' \hat{\Sigma}(a)^{-1} \hat{g}_n(a),$$

where

$$\hat{g}_n(a) = \frac{1}{n} \sum_{i=1}^n \left(\tau - \mathbf{1}\{Y_i \leq D_i a + X_i' \hat{\beta}(a)\} \right) \Psi(X_i, Z_i).$$

Here, $\hat{\beta}(a)$ is obtained by profiling out the nuisance coefficient for each candidate value a of the structural parameter.

The main advantage of Full-control IVQR is that it does not discard potentially relevant controls. It uses the full information contained in the control dictionary.

However, its main weakness is high-dimensional instability. When the number of controls p is large relative to the sample size n , estimating $\beta(\tau)$ can become noisy, unstable, or computationally difficult. This nuisance-estimation error may contaminate estimation of $\alpha(\tau)$. Therefore, Full-control IVQR is useful as a benchmark.

Post-Selection IVQR

The Post-selection IVQR estimator uses a two-step procedure.

First, a regularization method is used to select a subset of controls from the high-dimensional dictionary X . Denote the selected set by $\hat{S}_\tau \subset \{1, \dots, p\}$. The selected control vector is $X_{\hat{S}_\tau}$.

Second, IVQR is estimated using only the selected controls:

$$q_\tau(D, W) \approx \alpha(\tau)D + X_{\hat{S}_\tau}' \beta_{\hat{S}_\tau}(\tau).$$

The estimator is therefore

$$\hat{\alpha}_{PS}(\tau) = \arg \min_{a \in \mathcal{A}} \hat{g}_{PS,n}(a)' \widehat{\Sigma}_{PS}(a)^{-1} \hat{g}_{PS,n}(a),$$

where

$$\hat{g}_{PS,n}(a) = \frac{1}{n} \sum_{i=1}^n \left(\tau - \mathbf{1}\{Y_i \leq D_i a + X_{i,\widehat{S}_\tau}' \widehat{\beta}_{\widehat{S}_\tau}(a)\} \right) \Psi(X_{i,\widehat{S}_\tau}, Z_i).$$

The purpose of Post-selection IVQR is to reduce dimensionality before estimating the IVQR model. This may improve numerical stability and reduce overfitting relative to Full-control IVQR.

The main weakness is that the first-stage selection step can make mistakes. Relevant controls may be excluded, or irrelevant controls may be included. That is why, Post-selection IVQR is used as an intermediate estimator between Full-control IVQR and DML-IVQR.

DML-IVQR

The DML-IVQR estimator is the central estimator of the thesis.

Its purpose is to estimate the low-dimensional structural quantile treatment effect $\alpha(\tau)$ in the presence of high-dimensional nuisance parameters.

The key idea is to use an orthogonalized IVQR moment condition of the form

$$E[g_\tau(V, \alpha(\tau), \eta(\tau))] = 0,$$

where $V = (Y, D, Z, X)$ and $\eta(\tau)$ collects nuisance parameters such as $\beta(\tau)$ and additional components used to residualize or orthogonalize the instruments.

A generic DML-IVQR score can be written as

$$g_\tau(V, \alpha, \eta) = (\tau - \mathbf{1}\{Y \leq D\alpha + X'\beta\}) \Psi(\alpha, \delta),$$

where $\Psi(\alpha, \delta) = Z - \delta(\alpha)X$.

The nuisance parameter $\delta(\alpha)$ adjusts the instrument by removing the part that is explained by the controls. This produces an orthogonalized moment condition that is less sensitive to first-order errors in nuisance-parameter estimation.

The DML-IVQR estimator uses regularized methods to estimate the nuisance components. For each candidate value a of $\alpha(\tau)$, the nuisance coefficient $\beta(a)$ is estimated by penalized quantile regression:

$$\widehat{\beta}(a) = \arg \min_b \frac{1}{n} \sum_{i=1}^n \rho_\tau(Y_i - D_i a - X_i' b) + \lambda \sum_{j=1}^p \widehat{\sigma}_j |b_j|,$$

where $\rho_\tau(\cdot)$ is the quantile check loss function, λ is a penalty parameter, and $\widehat{\sigma}_j$ is a scale normalization factor.

The sample moment is

$$\hat{g}_{DML,n}(a) = \frac{1}{n} \sum_{i=1}^n \left(\tau - \mathbf{1}\{Y_i \leq D_i a + X_i' \hat{\beta}(a)\} \right) \hat{\Psi}_i(a),$$

where $\hat{\Psi}_i(a) = Z_i - \hat{\delta}(a)X_i$.

The DML-IVQR estimator is then obtained by minimizing a GMM-type criterion:

$$\hat{\alpha}_{DML}(\tau) = \arg \min_{a \in \mathcal{A}} n \hat{g}_{DML,n}(a)' \hat{\Sigma}_{DML}(a)^{-1} \hat{g}_{DML,n}(a).$$

The main advantage of DML-IVQR is that it combines three ideas:

1. regularization for high-dimensional nuisance estimation;
2. orthogonalized moment conditions to reduce first-order sensitivity to nuisance-estimation error;
3. weak-identification robust inference through a grid-search or GMM-type objective.

The expected benefit is that DML-IVQR should be more stable than Full-control IVQR when p is large and should be less sensitive to variable-selection mistakes than simple Post-selection IVQR.

Its weakness is computational complexity. It requires tuning choices, penalized quantile regression, instrument residualization, and repeated optimization over candidate values of $\alpha(\tau)$. Therefore, numerical stability must be evaluated explicitly in the Monte Carlo study.

4. Performance Measures

The main performance measures are: Bias, Median Bias, RMSE, MAE, Coverage, Confidence interval/region length, Non convergence rate.

5. Monte Carlo Design

DGP 1: Baseline Sparse Gaussian Design

For each Monte Carlo replication, observations are generated as

$$\{Y_i, D_i, Z_i, X_i\}_{i=1}^n \quad i = 1, \dots, n.$$

Here, Y_i is the outcome, D_i is the endogenous regressor, Z_i is the instrument, and X_i is a p -dimensional vector of controls.

The controls are generated as $X_i = (X_{i1}, \dots, X_{ip})'$, with $X_i \sim N(0, \Sigma_X)$. The covariance matrix Σ_X follows an autoregressive correlation structure:

$$(\Sigma_X)_{jk} = \rho_X^{|j-k|}, \quad \rho_X = 0.5.$$

This creates correlation among controls and makes the simulation more realistic than a design with fully independent covariates.

The instrument is generated as $Z_i \sim N(0, 1)$, independently of X_i and the structural disturbances.

The structural disturbances are generated as a bivariate normal vector with $\rho_{uv} = 0.5$:

$$\begin{pmatrix} u_i \\ v_i \end{pmatrix} \sim N\left(0, \begin{pmatrix} 1 & \rho_{uv} \\ \rho_{uv} & 1 \end{pmatrix}\right),$$

The correlation between u_i and v_i creates endogeneity. Since D_i depends on v_i , and v_i is correlated with u_i , the treatment variable is correlated with unobserved determinants of the outcome.

The outcome is generated as $Y_i = 1 + X_i'\beta + D_i(1 + u_i)$. This specification implies a structural quantile function of the form $q_\tau(D_i, X_i) = 1 + X_i'\beta + D_i\alpha_0(\tau)$, where the true structural quantile treatment effect is

$$\alpha_0(\tau) = 1 + \Phi^{-1}(\tau).$$

Therefore, for the three quantiles used in the thesis, $\tau \in \{0.25, 0.50, 0.75\}$, the true treatment effects are:

Quantile τ	True value $\alpha_0(\tau)$
0.25	approximately 0.326
0.50	1.000
0.75	approximately 1.674

This design creates heterogeneous structural quantile effects. The effect of D_i differs across the conditional outcome distribution, which is exactly the reason for using IVQR instead of mean-based IV methods.

DGP 2: Weaker Sparsity Design

DGP 2 keeps the same structure as DGP 1 but makes the nuisance component less sparse. This is done by increasing the number of relevant controls to $s = 20$.

The coefficient vectors are $\beta_j = \begin{cases} 0.5/\sqrt{j}, & j = 1, \dots, s, \\ 0, & j > s, \end{cases}$

and $\gamma_j = \begin{cases} 0.4/\sqrt{j}, & j = 1, \dots, s, \\ 0, & j > s. \end{cases}$

This design tests whether the estimators remain reliable when the approximate sparsity assumption becomes weaker.

DGP 3: Heavy-Tailed Error Design

DGP 3 keeps the same structural equations as DGP 1, but replaces Gaussian disturbances with heavy-tailed disturbances. The purpose is to test whether estimator performance deteriorates when extreme observations are more likely.

A simple implementation is to generate u_i and v_i from scaled Student- t disturbances with five degrees of freedom, while preserving positive correlation between them.

This design is used as a robustness check for the tail quantile results.

Simulation Scenarios

$n \in \{250, 500, 1000\}$, and $p \in \{200, 300, 500\}$.

$R = 1000$ Monte Carlo replications.

6. Estimators Computed in Each Replication

For each replication, quantile level, and simulation scenario, the following estimators are computed: $\hat{\alpha}_{Full}(\tau)$, $\hat{\alpha}_{PS}(\tau)$, $\hat{\alpha}_{DML}(\tau)$.

The estimators are compared to the true value $\alpha_0(\tau) = 1 + \Phi^{-1}(\tau)$.

The same simulated dataset is used for all three estimators within a given replication. This ensures that performance differences are due to estimator behavior rather than differences in simulated samples.

For the DML-IVQR estimator, nuisance components are estimated using K -fold cross-fitting with $K = 5$.

In each fold, the nuisance parameters are estimated using the observations outside the fold and then evaluated on the held-out fold. The fold-specific orthogonal moments are then combined to estimate $\alpha(\tau)$. Cross-fitting is used only inside the DML-IVQR estimator. Full-control IVQR and Post-selection IVQR are estimated on the full simulated sample without cross-fitting.

Performance Measures

Let $\hat{\alpha}_r(\tau)$ denote the estimate in replication r , and let $\alpha_0(\tau)$ denote the true structural quantile treatment effect. Then:

$$Bias(\tau) = \frac{1}{R} \sum_{r=1}^R (\hat{\alpha}_r(\tau) - \alpha_0(\tau)).$$

$$Median\ Bias(\tau) = median_{r=1, \dots, R} (\hat{\alpha}_r(\tau) - \alpha_0(\tau)).$$

$$RMSE(\tau) = \sqrt{\frac{1}{R} \sum_{r=1}^R (\hat{\alpha}_r(\tau) - \alpha_0(\tau))^2}.$$

$$\begin{aligned}
MAE(\tau) &= \frac{1}{R} \sum_{r=1}^R |\hat{\alpha}_r(\tau) - \alpha_0(\tau)|. \\
Coverage(\tau) &= \frac{1}{R} \sum_{r=1}^R \mathbf{1}\{\alpha_0(\tau) \in CI_r(\tau)\}, \\
Length_r(\tau) &= Upper_r(\tau) - Lower_r(\tau), \\
Average\ Length(\tau) &= \frac{1}{R} \sum_{r=1}^R Length_r(\tau). \\
Non-convergence\ Rate &= \frac{Number\ of\ failed\ replications}{R}.
\end{aligned}$$

7. Expected Results

High-Dimensional Controls

As the number of controls p increases relative to the sample size n , Full-control IVQR is expected to become less stable. This may appear through larger RMSE, wider confidence regions, or higher non-convergence rates.

Post-selection IVQR is expected to reduce dimensionality and improve stability, but it may suffer from selection mistakes.

DML-IVQR is expected to perform better in high-dimensional settings because it combines regularization with orthogonalized moments.

Weak Instruments

As the instrument-strength parameter π decreases, all IVQR-based estimators are expected to become less reliable.

Weak instruments may lead to: larger bias, RMSE and MAE; wider confidence regions; lower or unstable coverage.

Quantile Differences

Estimator performance may differ across quantiles. Tail quantiles, $\tau = 0.25$ and $\tau = 0.75$, are expected to be more difficult than the median, $\tau = 0.50$. This is because tail quantiles are more sensitive to extreme observations and may have weaker identification.

An estimator is considered better only if it has:

1. lower bias, RMSE, and MAE;
2. coverage close to the nominal level;
3. reasonably short confidence intervals or regions;
4. low non-convergence rate;